Homework Assignment 3

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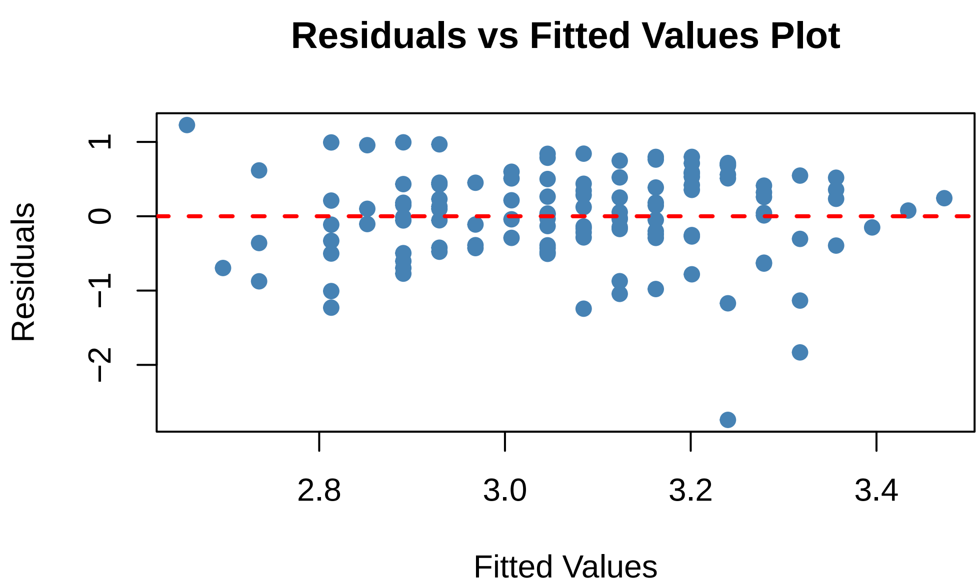
2026-04-28

Question 1 Refer to the GPA problem in the first two assignments. (a) (3 pt) Plot the residual against the fitted values. What departures from the linear regression model can be studied from this plot? What are your findings?

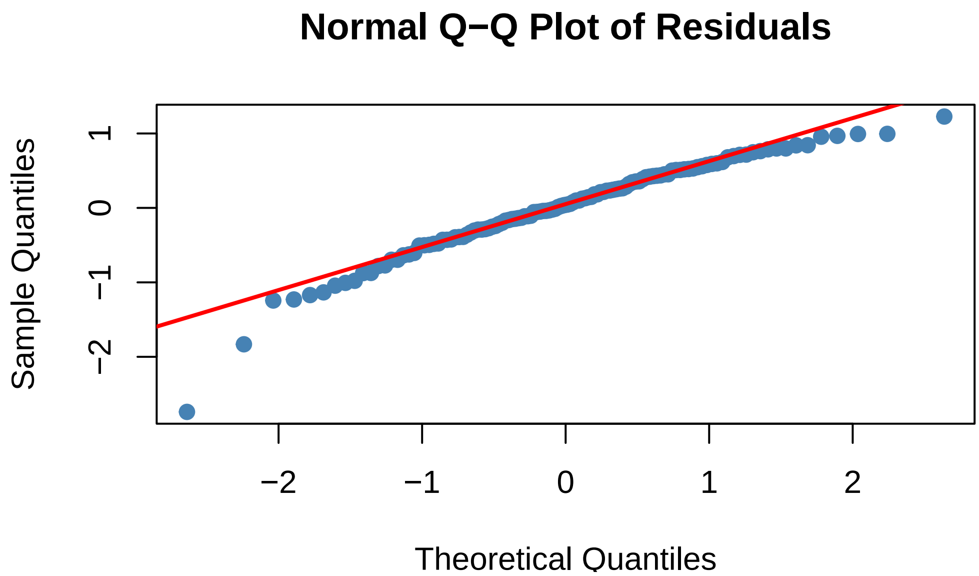
```r
gpa_data <- read.table("GPA.txt", header = TRUE)
gpa_model <- lm(GPA ~ ACT, data = gpa_data)
plot(gpa_model$fitted.values, gpa_model$residuals,
     xlab = "Fitted Values",ylab = "Residuals",
     main = "Residuals vs Fitted Values Plot",
     pch = 19, col = "steelblue")
abline(h = 0, col = "red", lwd = 2, lty = 2)
```



FRom this plot we can look at whether there is linearity which can be confirmed when the residuals are centered around zero. We can also confirm homoscedacity, or constant variance if there is not distict pattern or shape between the residuals. Lastly we are able to identify outliers. This plot shows the residuals roughly centered around zero without any distinct pattern that would look like homoscedacity Therefore this holds the linearity and constant variance assumption. There is very clear outliers such as the large residual by 3.2, but the model seems to be a good fit.

(b) (3 pt) Prepare a QQ plot of the residuals. What departures from the linear regression model can be studied from this plot? What are your findings?

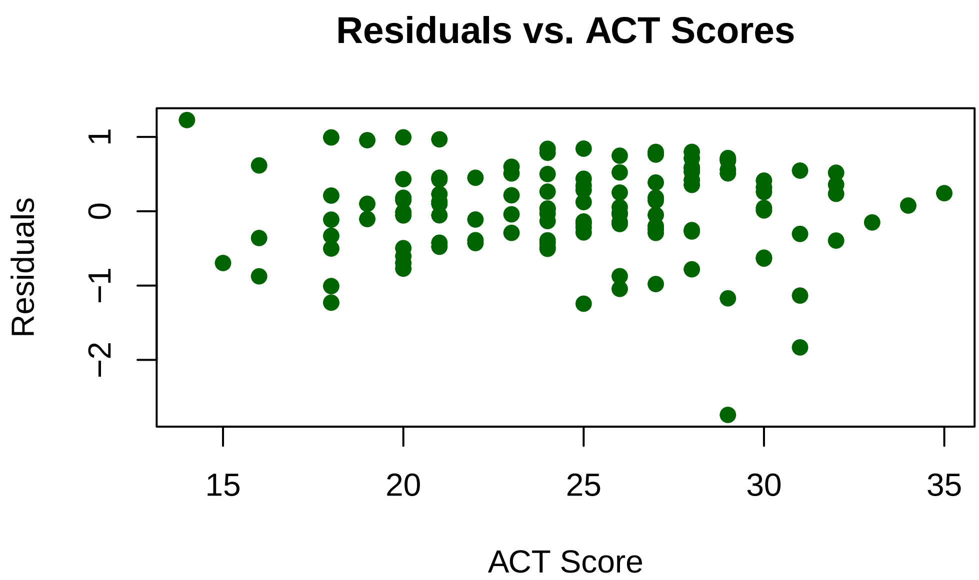
```r
res <- residuals(gpa_model)

qqnorm(res, main = "Normal Q-Q Plot of Residuals",
       xlab = "Theoretical Quantiles", ylab = "Sample Quantiles",
       pch = 19, col = "steelblue")

qqline(res, col = "red", lwd = 2)
```



The normal qq plot tells us whether the error terms follow a normal distribution. We find non normality is the points deviate from the qq line. We can also observe skewness, kurtosis and outliers. This qq plot seems to be approzimately normal because the straight line is followed closely. This indicated we can assume normality for the error term. We do see a significant outliers in the bottom left of the graph.

(c) (3 pt) Plot the residuals against the ACT scores. What departures from the linear regression model can be studied from this plot? What are your findings?

```r
plot(gpa_data$ACT, residuals(gpa_model),
     xlab = "ACT Score",ylab = "Residuals",
     main = "Residuals vs. ACT Scores",
     pch = 19, col = "darkgreen")
```

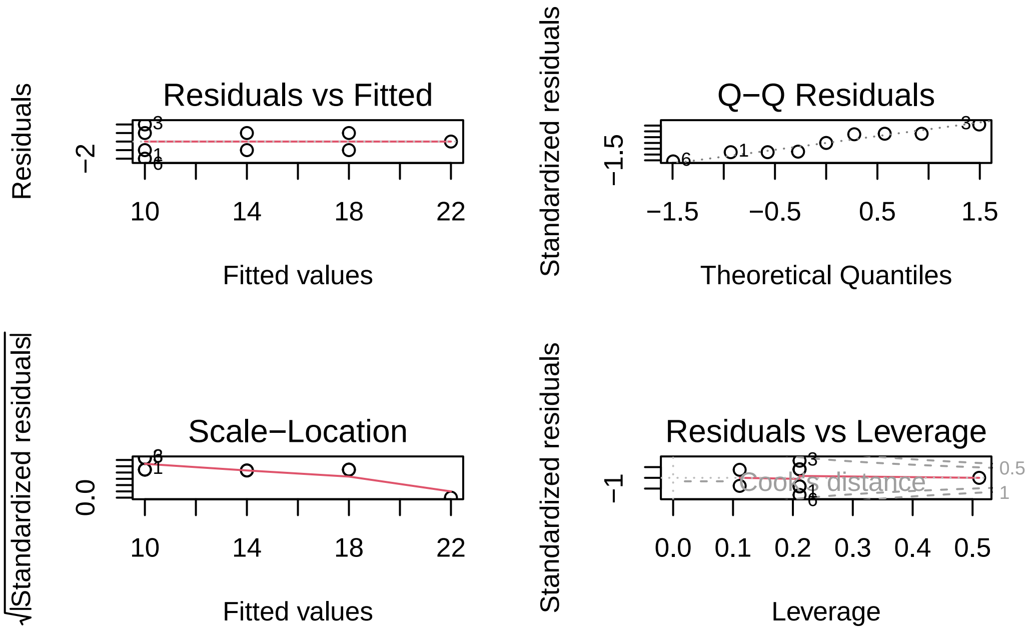


The departures from the linear regression model are Non-linearity-> if the residuals form a parrter such as a curve it can suggust a non linear pattern, might need a log transform Heteroscedasiticity -> non constant variance if the sprad of residuals differes based on act score Outliers From this plot, there is a consistent linear relationship as the residuals are scattered randomly around the 0 line and there is no obvious curve. There is uniform variance and the spread of residuals does not seem to change as act scores increase. We still observe the same outlier

(d) (3 pt) Discuss possible remedial measures when there are departures in (a), (b) and (c) If the data schows to be non linear in these plots, we could apply a log transform or square root transform to linearize the graph. If we see non constant variance we also do a log transform. Sometimes we can also remove outliers is they seem to be a data entry error or misrepresenting the population

2. (10 pt) Refer to the Airfreight breakage problem in the first two assignments. Construct diag- nostic plots and comment on your findings.

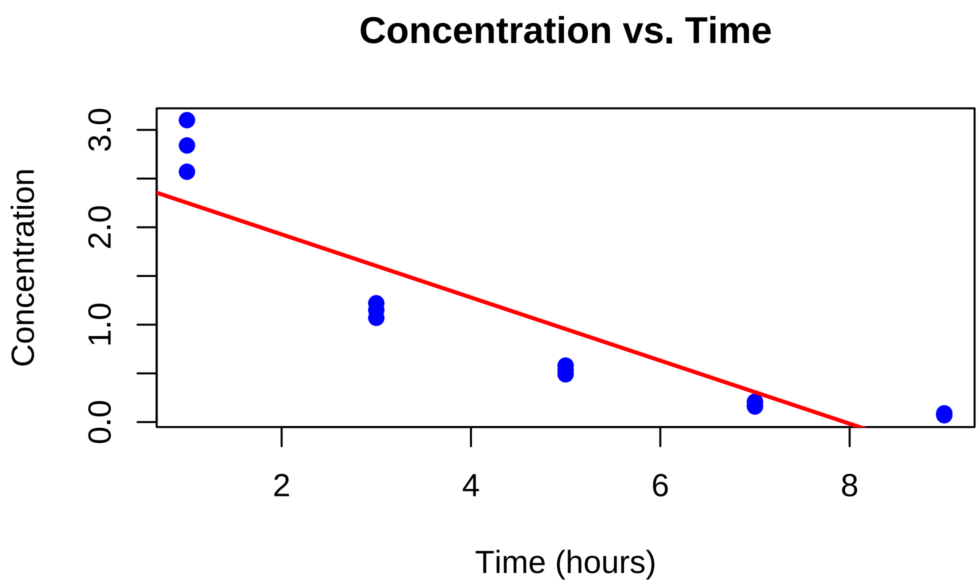
```r
air_data <- read.table("airfreight_breakage.txt", header = TRUE)
colnames(air_data) <- c("broken", "transfers")
fit <- lm(broken ~ transfers, data = air_data)

par(mfrow = c(2, 2))
plot(fit)
```



Residuals vs Fitted The residuals show a slight megaphone shape around the line. The variance still may be consistent just because the points still do look random and we only have a couple observations to go off of QQ Residuals The points lie closely to the line meaning that the error terms seem approzimately normal Scale-Location This line isn't perfectly straight however it only changes by 1.2 standardized residuals which seems minimal given the small sample size. this leads me to assume that the variance is constant Residuals vs Leverage All points fall within the boundaries, meaning no point is overly influential The model, in my opinion, is a decently good fit. It satisfies assumptions for linarity, normality and constant variance

Question 3 A chemist studied the concentration of a solution over time. Fifteen identical solutions were prepared. The 15 solutions were randomly divided into five sets of three, and the five sets were measured, respectively, after 1, 3, 5, 7, and 9 hours. A dataset named "Solution data" is available at the Canvas. In this problem, time is the independent variables and concentration is the response variable. (a) (3 pt) Plot the data and fit a linear regression model.

```r
solution_data <- read.table("solution.txt", header = TRUE)
plot(solution_data$time, solution_data$concentration,
     xlab = "Time (hours)",
     ylab = "Concentration",
     main = "Concentration vs. Time",
     pch = 19, col = "blue")
conc_model <- lm(concentration ~ time, data = solution_data)

abline(conc_model, col = "red", lwd = 2)
```



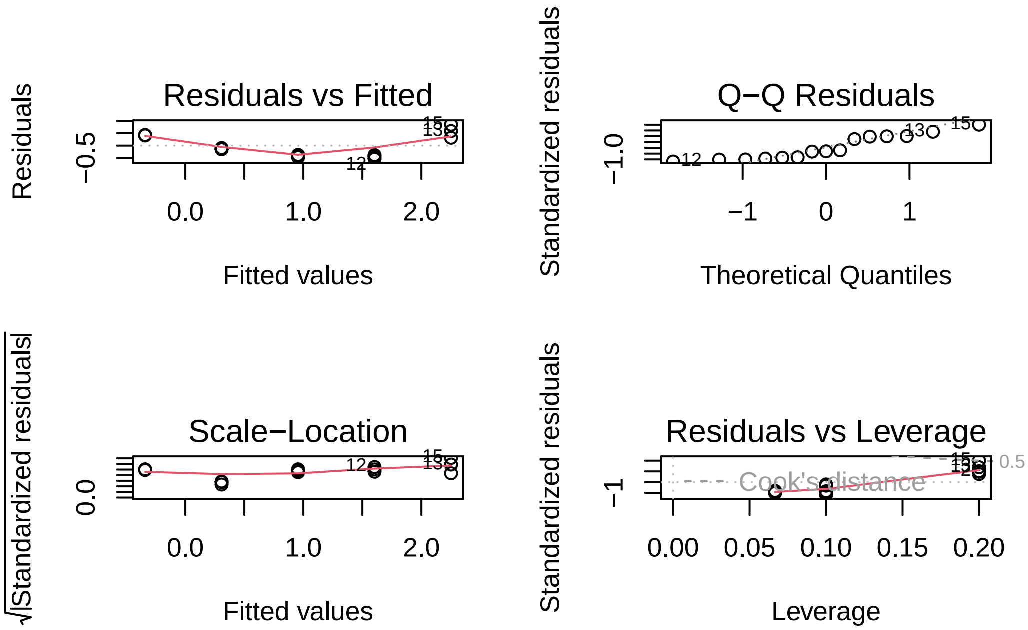

```
summary(conc_model)
```

```
Call:
lm(formula = concentration ~ time, data = solution_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.5333 -0.4043 -0.1373  0.4157  0.8487 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   2.5753     0.2487  10.354 1.20e-07 ***
time         -0.3240     0.0433  -7.483 4.61e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4743 on 13 degrees of freedom
Multiple R-squared:  0.8116,	Adjusted R-squared:  0.7971 
F-statistic: 55.99 on 1 and 13 DF,  p-value: 4.611e-06
```

Concentration = 2.5753 - .324(time) The p-value is very small and significant so we can conclude there is a linear relatinoship The multiple R^2 tells us 81.16% of the variation in concentration is explained by time

b) (3 pt) Construct diagnostic plots and comment on your findings.

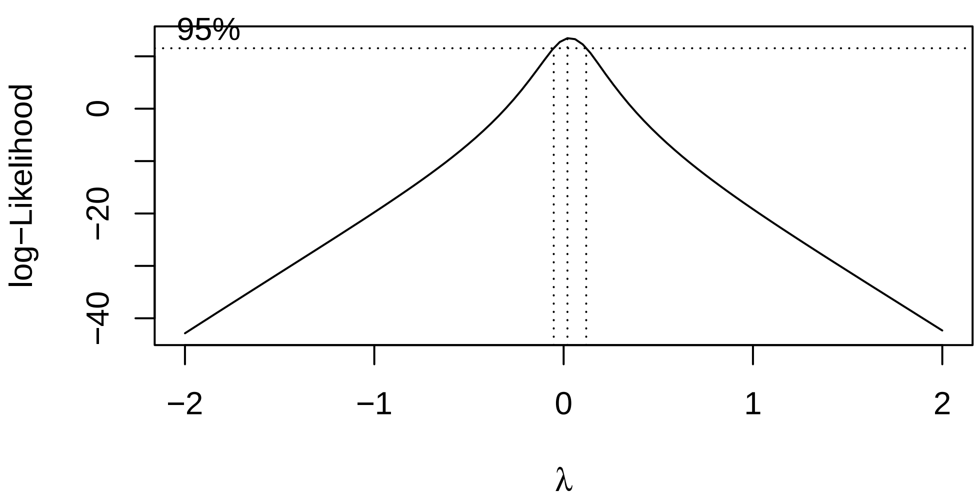
```
par(mfrow = c(2, 2))
plot(conc_model)
```



The residuals vs fitted has a slight U shape but only deviates by 1 so it may not be linear, it is hard to definitively judge because the sample is so small. The QQ plot checks normality, and residuals fall along the line showing the errors are appozimately normal. The scale locatino plot suggusts constant variance because the fitted values are equaly close to the line The residuals vs leverage shows all points falling within the boundaries and none being overly influential

(c) (3 pt) Use the Box-Cox procedure to find an appropriate power transformation.

```r
library(MASS)

bc <- boxcox(conc_model)
```




```r
(lambda <- bc$x[which.max(bc$y)])
```

```
[1] 0.02020202
```

Since lambda is close to zero, I will use a log transform

(d) (3 pt) Use the transformation Y = log10(concentration) and obtain the estimated linear regression function for the transformed data.

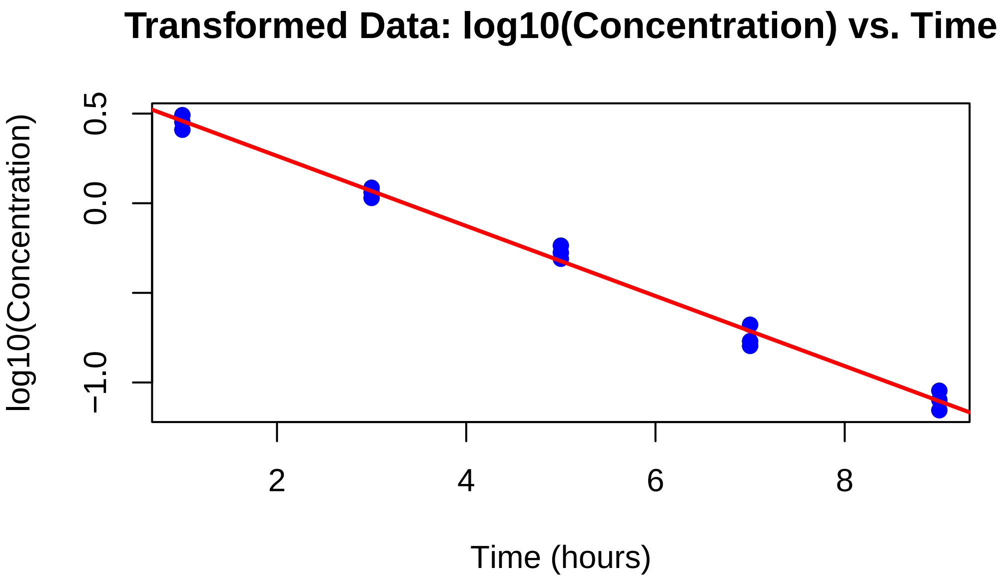
```r
conc_model_log <- lm(log10(concentration) ~ time, data = solution_data)
summary(conc_model_log)
```

```

Call:
lm(formula = log10(concentration) ~ time, data = solution_data)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.082958 -0.044421  0.006813  0.033512  0.085550 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.654880   0.026181   25.01 2.22e-12 ***
time        -0.195400   0.004557  -42.88 2.19e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04992 on 13 degrees of freedom
Multiple R-squared:  0.993,	Adjusted R-squared:  0.9924 
F-statistic:  1838 on 1 and 13 DF,  p-value: 2.188e-15
```

log10(Concentration) = .65488 - 0.1954(time)

(e) (3 pt) Plot transfermed data and the estimated regression line in (d). Does the regression line appear to be a good fit to the transformed data?

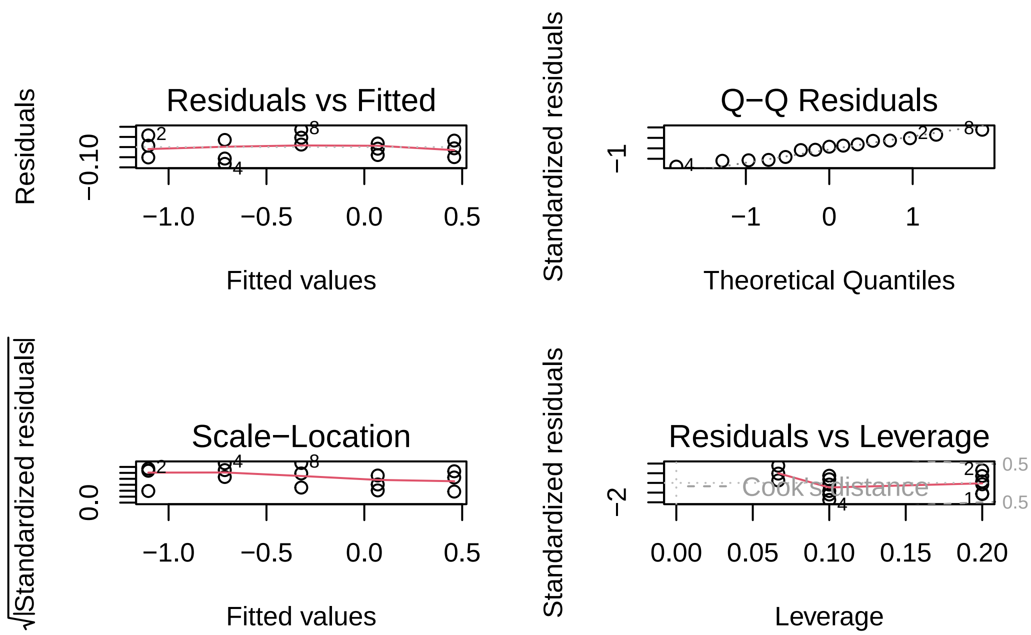
```r
plot(solution_data$time, log10(solution_data$concentration),
     xlab = "Time (hours)",
     ylab = "log10(Concentration)",
     main = "Transformed Data: log10(Concentration) vs. Time",
     pch = 19, col = "blue")
abline(conc_model_log, col = "red", lwd = 2)
```



The regression line appears to be an awesome fit and the data points fall almost exactly on the line.

(h) (3 pt) Construct diagnostic plots for the fit in (d) and comments on your findings.

```r
par(mfrow = c(2, 2))
plot(conc_model_log)
```



The residuals vs fitted is scattered and has no curve showing a linear relationship between time and concentration now the qq plot points follow the line closely showing that the error terms follow a normal distribution the scale location plot has a flat trend line telling us that the variance is now stabalized and the residuals vs leverage has all points within the boundaries

(i) (3 pt) Express the estimated regression function in (d) in the original units.

log10(Concentration) = .65488 - 0.1954(time) Concentration = 10^(.65488 - 0.1954(time)) Concentration = 4.5173 * 0.6377^Time